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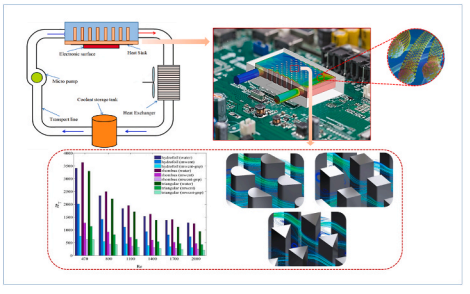
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Irreversibility and hydrothermal analysis of the MWCNTs/GNPs-based nanofluids for electronics cooling applications of the pin-fin heat sinks: Multiphase Eulerian-Lagrangian modeling

Tehmina Ambreen [a], Arslan Saleem [a], Muhammad Tanveer [a], Anirudh K [b], S. A. Shehzad [c], Cheol Woo Park [a,*]

[a] School of Mechanical Engineering, Kyungpook National University, 80 Daehakro, Bukgu Daegu, 41566, South Korea
[b] Department of Mechanical Engineering, School of Technology, Pandit Deendayal Energy University, Raisan, Gandhinagar, 382 426, India
[c] Department of Mathematics, COMSATS University Islamabad, Sahiwal, 57000, Pakistan

GRAPHICAL ABSTRACT



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ABSTRACT

Identifying optimal pin-fin configurations is essential to analyze the full potential of utilizing nanofluids in pin-fin heat sinks for electronics cooling applications. Therefore, the present study aimed to investigate the entropy generation and hydrothermal characteristics of water-based MWCNT (Multiwalled carbon nanotubes) and MWCNT/GNP (Multiwalled carbon nanotubes/graphene nanoplatelets) nanofluids in a heat sink equipped with fins of different configurations. The forced convection of the nanofluid was simulated through the multiphase Eulerian-Lagrangian model (discrete-phase model) by including the influence of slip mechanisms (Brownian motion, thermophoresis, Saffman lift and drag forces, pressure forces, virtual mass, and gravity) and the interphase nanolayering effect. Results showed that the performance of the MWCNT and MWCNT/GNP nanofluids was the highest in the triangular pin-fin heat sink,

* Corresponding author.
E-mail address: chwoopark@knu.ac.kr (C.W. Park).

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followed by rhombus and hydrofoil pin-fin heat sinks. The streamlined shape along with the large lateral and small stream-wise interspacing between the hydrofoil and rhombus fins hindered the effective coolant distribution across the heat sinks, thereby deteriorating their heat-dissipation capacity. However, the performance evaluation criteria results of the rhombus pin-fin heat sink outperformed the other pin-fin configurations due to the lower pressure drop.

Nomenclature

Al_2O_3 Aluminium oxide (-)
C_c Cunningham correlation (-)
C_f Inertial Coefficient (-)
C_D Drag Coefficient (-)
C_U Copper (-) ($Jkg^{-1}K^{-1}$)
C_P Specific heat ($Jkg^{-1}K^{-1}$)
d Diameter (m)
d_{ij} Deformation tensor (s^{-1})
D_h Channel hydraulic diameter (m)
F_{lift} Lift force (ms^{-2})
F_{drag} Drag force (ms^{-2})
$F_{gravity}$ Force owing to gravity (ms^{-2})
$F_{virtual}$ Virtual mass force (ms^{-2})
$F_{thermophoresis}$ Thermophoresis force (ms^{-2})
$F_{bownian}$ Brownian force (ms^{-2})
$F_{Pressure}$ Pressure force (ms^{-2})
Fe_3O_4 Iron oxide (-)
EG Ethylene glycol (-)
GNPs Graphene nanoplatelets (-)
g Gravitation acceleration (ms^{-2})
H Height (m)
H_f Fin height (cm)
H_t Heat sink height (cm)
h Convective heat-transfer coefficient ($W\ m^{-2}K^{-1}$)
H_2O Water (-)
Kn Knudson number (-)
k Thermal conductivity ($Wm^{-1}K^{-1}$)
Nu Nusselt number (-)
m Mass (kg)
M Molar mass ($kg.mol^{-1}$)
MWCNT Multiwalled carbon nanotubes (-)
P Pressure (Nm^{-2})
PEC Performance evaluation criteria (-)
K_B Boltzmann constant (JK^{-1})
L Length (m)
L_f Length of fin occupied area (cm)
L_t Heat sink length (cm)
LED Light-emitting diode (-)
N Avogadro's number (-)
Nt Number of nodes (-)
P_r Prandtl number (-)
Q Heat flux (kWm^{-2})
Q" Local heat flux (Wm^{-2})
Re Reynolds number (-)
T Temperature (K)
V Velocity (ms^{-2})
W_f Width of fin occupied area (cm)
W_t Heat sink width (cm)
S Entropy generation ($Wm^{-3}K^{-1}$)
S' Global entropy generation rate (WK^{-1})
S_m Momentum source term (-)

S_e Energy source term (-)
SiO_2 Silicon Oxide (-)
s_fFin interspacing (cm)
TTemperature (K)
TiO_2 Titanium dioxide (-)
t Time (s)
V Velocity (m s^{-1})
W Width
Δy Near wall element size (mm)
Y + Wall distance of initial node (-)
ZnO Zinc Oxide

Greek Symbols
μ Viscosity ($kgm^{-1}s^{-1}$)
β Nanoparticle aspect ratio (-)
$\overline{\tau}$ Stress-strain tensor (-) Density (kgm^{-3})
ψ Particle weight fraction (-)
φ Particle effective volume fraction (-)
η_r Relative viscosity of nanofluid (-)
v Kinematic viscosity (ms^{-2})

Subscripts
a Average
am Ambient
f fin
fr viscous
in Heat sink inlet
nf Nanofluid
n Nanoparticles
r Relative
s Nanoparticle phase
m Heat sink material
T Total
th Thermal
T Reference temperature
out Heat sink outlet
w Wall

1. Introduction

Convective heat transfer, a vital part of energy-utilization processes, offers considerable potential for energy conservation. Innovative methodologies and thermal fluids with the promise of optimizing the heat-transfer rate are keys to ensuring efficient cost-effective energy solutions [1]. In this pursuit, nanofluids synthesized by stably suspending nanosized metallic/nonmetallic entities in conventional fluids are a possible candidate [2–5]. In the past two decades, an immense amount of research has been carried on nanofluids considering the thermophysical properties, synthesis methods, computational methods and hydrothermal characteristics, magnetic properties, and magnetic force-driven flow behavior [6–13]. The hydrothermal performance of the nanofluids is governed by several factors such as nanoparticle concentration, morphology, and thermophysical properties, operational temperature, and thermophysical properties of the hosting fluid. Generally, the performance of the nanofluids improves at larger particle concentration, reduction in particle size, increment in particle effective surface area, and thermal conductivity [14–16]. Among different nanofluid combinations, nanofluids based on MWCNTs (multiwalled carbon nanotubes) and GNPs (graphene nanoplatelets) are highly efficient thermal fluids because of their adjustable properties and ultrahigh thermal conductivity of the carbon nanomaterial (up to 5000 W/mK) [17].

Ding et al. [18] performed a preliminary investigation on MWCNT based nanofluid and demonstrated a maximum convective heat-transfer enhancement of 350% by utilizing MWCNT-H_2O nanofluid with particle concentrations of 0.5%. Later, several experimental and numerical studies highlighted the preparation and characterization techniques, thermophysical properties, and hydrothermal performance of MWCNT nanofluids [19]. A few researchers have introduced the concept of hybrids of MWCNT nanofluid with other metallic and nonmetallic nanoparticles to obtain desirable characteristics [20]. Sundar et al. [21,22] examined the hydrothermal features of MWCNT/Fe_3O_4-H_2O (0.1 and 0.3 vol%) hybrid nanofluid in a tube with and without tape (twisted and longitudinal) inserts and reported the maximum Nusselt number improvement of 50.99% at the expense of a 26% increase in pressure drop. Ahmed et al.

[23] analyzed the forced convection of water-based nanofluid comprising MWCNTs and GNPs nanoparticles. The dispersion of hybrid nanoparticles results in the maximum heat transfer and pressure-drop augmentation of 43.4% and 11%, respectively. Kumar and Sarkar [24] evaluated the performance of an Al_2O_3/MWCNT-H_2O (0.3 vol%) nanofluid in a minichannel heat sink. The optimum ratio of the heat transfer and pressure drop was obtained with a 3:2 ratio of Al_2O_3 to MWCNTs. Recently, Alnaqi et al. [25] analyzed the hydrothermal performance analysis of the MWCNT/SiO_2-EG-H_2O nanofluid in a zigzag microchannel.

Pin-fin heat sinks are used for a wide range of cooling applications such as electronic equipment, LED chips, semiconductor lasers, and microreactors. The heat sinks equipped with mini- and micro-pin-fins have evolved in recent years to optimize the heat exchange rate by augmenting the surface-to-volume ratio and uniform temperature distribution [26–30]. Additionally, disruption of the viscous-thermal boundary layer, intensive coolant mixing, and low thermal resistance of micropin-fin heat sinks propose a compact solution for electronic thermal management. The hydrothermal performance of these heat sinks primarily depends on fin arrangement, morphology, porosity, tip clearance, and coolant distribution [31–38]. Turkyilmazoglu [39] demonstrated that a shrinking fin surface demonstrates an improved fin efficiency and heat transfer rate. Several studies have demonstrated the performance enhancement of micropin-fin heat sinks by utilizing different types of nanofluids [40]. Duangthongsuk and Wongwises [41] examined the hydrothermal features of a miniature pin-fin (circular) heat sink by utilizing SiO_2/H_2O and ZnO/H_2O nanofluid and demonstrated the superior thermal efficiency of ZnO/H_2O nanofluid compared with that of SiO_2/H_2O nanofluid at all particle loadings ($0.2\% \leq \varphi \leq 0.6\%$). Zhou et al. [42] experimentally analyzed the fluid flow and convective heat transfer of silver nanofluids cooled micro pin-fin heat sink. Ali and Arshad [43] experimentally assessed the thermal performance of the TiO_2 and GNP-based aqueous nanofluids by considering the influence of fin arrangements (inline and staggered) and fin angles (22.5°–90°), respectively. Ambreen et al. [44–46] probed the hydrothermal efficacy of different nanofluid combinations over the inline and staggered pin-fin assembly and emphasized that the pin-fin configuration significantly affects the wake structures behind the pin-fins and, the viscous and thermal boundary layers. Fatima and Mohamed [47] analyzed the mixed convective heat transfer of different metallic nanofluids in a heat sink equipped with perforated pin-fins of circular cross-sections. Mehdi et al. [48] simulated the forced convection of graphene nanoplatelets based nanofluids in miniature pin-fin heat sink in the homogenous approximation and concluded that pin-fins of streamlined shape improve the heat dissipation capacity of the heat sink. Yulin Ma et al. [49] evaluated the influence of the pin-fin assembly on the performance of a bifurcation micro-channels heatsink operating on the Ag/H_2O nanofluids by utilizing the two-phase mixture model.

It is clear from the literature survey that identifying suitable pin-fin configurations is essential to realize the full potential of utilizing nanofluids in enhancing the hydrothermal performance of pin-fin heat sinks. Therefore, the present study aimed to investigate

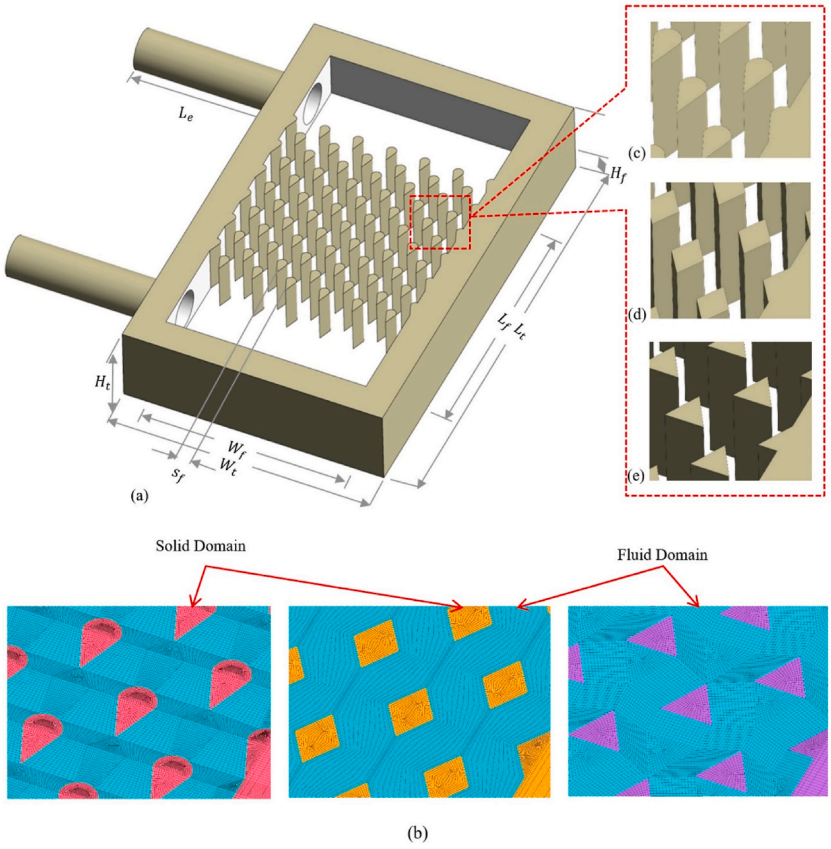


Fig. 1. (a) Schematic and (b) computational grid of the heat sinks with (c) hydrofoil, (d) rhombus, and (e) triangular fins.

the irreversibility and hydrothermal analysis of aqueous-based MWCNT and MWCNT/GNP nanofluids in a heat sink with variable pin-fin configurations, due to the high thermal conductivity of these nanofluids [17]. The heat sinks consisted of stagger pin-fin assembly of hydrofoil, rhombus, and triangular pin-fins. The key objective of the study was to assess the cooling performance and hydraulic characteristics of the MWCNT and MWCNT/GNP nanofluids dependent to fin types. The forced convection of a nanofluid was simulated through the multiphase discrete-phase model by incorporating the influence of slip mechanisms and interphase nano-layering effect. No such investigation has been found in the published literature to the best of the authors' knowledge.

2. Methodology

2.1. Heat sink schematic

The modeled computational domain comprised a pin-fin heat sink of aluminum material equipped with the staggered arrangement of different hydrofoil-shaped pin-fins ($H_f = 0.25$ cm, $s_f = 0.083$ cm). The length (L_t), width (W_t) and height (H_t) of the heat sink are 1.8 cm, 1.3 cm, and 0.4 cm, respectively as depicted in Fig. 1(a). Constant heat flux was implemented onto the heat sink base underneath the pin-fin-occupied area ($L_f \times W_f = 1$ cm $\times$ 1 cm). The coolant entering from the inlet flowed over the pin-fins and dissipated heat from the base of the heat sink through convection before exiting from the outlet. The nanoparticle mass fraction was 0.25% for the MWCNT-H_2O nanofluids, while an additional 0.035% mass fraction of the GNPs nanoparticles were introduced in each sample of the MWCNT/GNPs-H_2O nanofluids [23]. The pin-fins were designed in such a manner that the total wet surface area of the fins (A_f) and heat sink base (containing fins) were constant for all heat-sink configurations.

The study operational conditions are enlisted in Table 1.

2.2. Lagrangian–Eulerian (L-E) model

The two-phase approaches to simulate the forced convection of the nanofluids are generally classified as Eulerian–Eulerian and Lagrangian-Eulerian based on the treatment of nanoparticles and liquid phases. The Eulerian–Eulerian approximation (Volume of fluid, Eulerian, and mixture models) of the nanofluids is based on the assumption of the continuous inter-penetration of the particle and the liquid phases. While in the Lagrangian-Eulerian approach, the nanoparticles are treated as discrete entities and the trajectories of the nanoparticles (MWCNT and GNPs) are traced in a Lagrangian approach, whereas the primary phase (water) governing equations are simulated by implementing continuum assumption. The interphase (liquid and granular phases) momentum and energy exchange was computed by defining source terms into the liquid phase transport equations. The momentum interactions comprised of nanoparticle Brownian motion, thermophoresis, Saffman lift and drag forces, pressure forces, virtual mass, and gravity.

In this study, the simulation of the forced convection of nanofluids was performed in Lagrangian-Eulerian approximation due to the higher accuracy of this approach as highlighted by past literature studies [50–54]. As the nanoparticle concentration was very small in the nanofluid solution, therefore, the interparticle collisions were assumed as negligible. The transport equations of the primary phase are given as [45];

$$\nabla \cdot \left(\rho_f \vec{V}_f \right) = 0 \tag{1}$$

$$\nabla \left(\rho_f \vec{V}_f \vec{V}_f \right) = -\nabla P + \nabla . \left(\mu_f \nabla \vec{V}_f \right) + S_m \tag{2}$$

$$\nabla . \left(\vec{V}_f \rho_f C_{p_f} T_f \right) = \nabla . \left(k_f \nabla T_f \right) + S_e \tag{3}$$

where

$$S_m = \sum n_s \frac{m_s}{\delta \mathscr{V}} \frac{dV_s}{dt} \tag{4}$$

$$\frac{dV_s}{dt} = F_{drag} + F_{gravity} + F_{Brownian} + F_{thermophoresis} + F_{lift} + F_{virtual} + F_{pressure} \tag{5}$$

where F_{lift} and F_{drag} are estimated using the Saffman lift force [55] and Stokes drag [56] correlations, respectively.

Table 1
Study operational conditions.

Parameter/Operational Condition	Specification
Reynolds number, Re	470–2000
Heat flux, Q	3000 kW/m^2
Heat sink material thermal conductivity (aluminum), k_m	202.4 W/m-k
Ambient temperature, T_{am}	300 K
Ambient pressure, P_{am}	101.325 kPa
Nanoparticle concentration (volume fraction), φ	0.25%

$$F_{lift} = \frac{5.188 v_f^{0.5} \rho_f d_{ij}}{\rho_s d_s (d_{lk} d_{kl})^{0.25}} (V_f - V_s) \tag{6}$$

$$F_{drag} = \frac{18 \mu_f}{d_p^2 \rho_p C_c} (V_f - V_s) \tag{7}$$

The forces browning motion ($F_{Brownian}$) [57], thermophoresis ($F_{thermophoresis}$) [58], gravity ($F_{gravity}$), virtual mass ($F_{virtual}$) and pressure gradient forces ($F_{pressure}$) [59] are simulated using the following equations:

$$F_{Brownian} = \zeta_i \sqrt{\frac{\pi}{\Delta t} \frac{216 \nu_f K_B T}{\pi^2 \rho_f d_p^5 \left(\frac{\rho_s}{\rho_f}\right)^2 C_c}} \tag{8}$$

$$F_{thermophoresis} = \frac{7.02 \pi d_s \mu_f^2 (K + 2.18 K_n)}{m_s \rho_f T (1 + 3.42 Kn)(1 + 2K + 4.36 Kn)} \times \frac{\partial T}{\partial x} \tag{9}$$

$$F_{gravity} = \frac{g(\rho_s - \rho_f)}{\rho_s} \tag{10}$$

$$F_{virtual} = 0.5 \frac{\rho_f}{\rho_s} \frac{d}{dt} (V_f - V_s) \tag{11}$$

$$F_{pressure} = \left(\frac{\rho_f}{\rho_s}\right) V_s \frac{dV_f}{dx} \tag{12}$$

In Eq. (9), the $K = \frac{k}{k_s}$ represents the deformation tensor and $Kn = \frac{2\lambda}{d_s}$ defines the Knudsen number. The ζ_i in Eq. (8) signifies the zero-mean and unit-variance-independent Gaussian random numbers, and C_c represents the modified Cunningham correction factor [56] for the non-spherical particles given as [60]:

$$C_c = \frac{3 d_L [\ln(2\beta) - 0.72]}{2} \tag{13}$$

In the above equations, β denotes the aspect ratio of the particle length to diameter.
In Eq. (3), the energy-source term S_e is defined as follows [59]:

$$S_e = \sum n_s \frac{m_s}{\delta \mathscr{V}} C_p \frac{dT_s}{dt} \tag{14}$$

$$m_s C_s \frac{dT_s}{dt} = h_i A_s (T_f - T_s) \tag{15}$$

$$Nu = \frac{h_i d_s}{K_f} = 2 + 0.6 Re_s^{0.5} Pr^{0.33} \tag{16}$$

where Nu values are assessed through the Ranz and Marshall model [61]. To simulate the forced convection of base fluid (water) without the particulate phase, the governing Eqs. (1)–(3) are solved without the momentum and energy-source terms. For the case of conduction through the solid domain, the energy equation for conduction is given as follows [62–64]:

$$k_s \left(\frac{\partial^2 T_s}{\partial x^2} + \frac{\partial^2 T_s}{\partial y^2} + \frac{\partial^2 T_s}{\partial z^2}\right) = 0 \tag{17}$$

The density (ρ_{nf}), temperature-dependent viscosity (μ_{nf}) and thermal conductivity (k_{nf}) of the nanofluids are estimated by Eq. (18) [65], 19 [66], and 20 [67] respectively.

$$\rho_{nf} = \varphi \rho_s + (1 - \varphi) \rho_f \tag{18}$$

$$\mu_{nf} = (1 + 2.5\varphi + 6.2\varphi^2) \mu_f(T) \tag{19}$$

$$k_{nf} = \left(\frac{k_s + (n-1) k_f(T) + (n-1)(k_s - k_f(T))\varphi}{k_s + (n-1) k_f(T) - (k_s - k_f(T))\varphi}\right) k_f\left(T\right) \tag{20}$$

where [67];

$$\varphi = \frac{\psi \rho_f}{\rho_s + \psi \rho_f - \psi \rho_s} \tag{21}$$

The effective volume fraction of the nanofluids φ is estimated by using the following expression [68].

$$\varphi = \frac{\eta_r - 1}{2.5} \tag{22}$$

In the above equation, the η_r is the relative viscosity which defines the ratio of nanofluid solution viscosity to its base fluid viscosity ((μ_{nf}/μ_f).

The thermophysical properties (ρ_s, k_s C_{Ps}) of the nanoparticles are given in Table 2. Whereas, the temperature-dependent thermophysical properties (ρ_f, k_f, C_{Pf}, μ_f) of the base fluid are determined by using Eqs. (23)–(26) [69] as;

$$\rho_f = 2446 - 20.674T + 0.11576\ T^2 - 3.12895 * 10^{-4}T^3 - 2.0546 * 10^{-10}T^5 \tag{23}$$

$$\left(C_p\right)_f = \exp\left(\frac{8.29041 - 0.012557T}{1 - \left(1.52373*10^{-3}\right)T}\right) \tag{24}$$

$$\mu_f = 2.414*10^{-5}*10^{\left(\frac{247.8}{T-140}\right)} \tag{25}$$

$$k_f = 0.5706 + 1.756*10^{-3}\ T\ - 6.46*10^{-6}\ T^2 \tag{26}$$

2.3. Boundary conditions

The heat sink inlet and outlet were assigned with the velocity inlet and pressure outlet boundary conditions, respectively. An escape boundary condition was also implemented at the inlet and outlet of the heat sink to trace the particulate phase. The heat sink was assumed to be insulated from all sides except at the base of the heat sink below the fin occupied surface, which was assigned with the constant heat-flux boundary condition. All the inner walls and interfaces were specified with the no-slip and reflect (for particulate phase tracing) boundary conditions. The coolant temperature and pressure at the heat-sink inlet were presumed to be equivalent to the ambient conditions.

2.4. Numerical procedure

A three-dimensional numerical solution of the problem under consideration was obtained by utilizing FVM (finite volume method) based commercial software. The numerical details are similar to the past study of the author based on the comparative evaluation of the single and multiphase numerical models to simulate forced convection of MWCNTs/GNPs-based nanofluid [60]. A coupled SIMPLE algorithm was utilized to handle the pressure–velocity coupling. All the terms were interpolated by using the second-order-upwind scheme except pressure terms which were discretized using the standard discretization scheme. For each simulation, initially, the transport equations of the liquid phase (Eqs. (1)–(3) without source terms) were computed for obtaining a converged flow field. The discrete phase and the related interphase interactive forces were initiated in the following step. The nanoparticles were introduced by surface injection, which is normal for heat-sink inlet surfaces, and their unsteady path was traced by the combined low-order-implicit and high-order-trapezoidal schemes. The tolerance of the accuracy control limit for particle tracking was 10^{-6}. While the convergence criteria to simulate all the transport equations were restricted below 10^{-5} to limit the maximum mass residual of the grid control volume.

2.5. Performance parameters

The hydrothermal-performance parameters of the heat sink were estimated in terms of local Nusselt number (Nu_{local}), average Nusselt number enhancement ratio ($Nu_{a,nf}/ Nu_{a,f}$), pressure-drop enhancement ratio ($\Delta P_{nf}/ \Delta P_f$), and performance-evaluation criteria (PEC). Nu_a and Nu_{local} were determined by the following expressions;

$$Nu_a = \frac{h_a S_f}{k_{f,nf}} \tag{27}$$

$$Nu_{local} = \frac{h_s s_f}{k_{f,nf}} \tag{28}$$

where the local (h_{local}) and average (h_a) convective heat-transfer coefficients were determined as follows:

Table 2
Thermophysical properties of MWCNTs and GNPs.

Material	ρ[kg m^{-3}]	Cp[J kg^{-1}K^{-1}]	k[W m^{-1}K^{-1}]	Ref.
MWCNT	2100	630	1500	[70]
GNP	2200	790	3000	[23]

$$h_a = \frac{\left(h_f A_f - h_{bs} A_{bs}\right)}{\left(A_f + A_{bs}\right)} \tag{29}$$

$$h_{local} = \frac{Q^{''}}{T(z)_{bs} - T(z)_m} \tag{30}$$

The PEC was calculated as follows:

$$PEC = \frac{Nu_{a,nf} / Nu_{a,f}}{\left(\frac{f_{nf}}{f_f}\right)^{1/3}} \tag{31}$$

where

$$\frac{f_{nf}}{f_f} = \frac{\Delta P_{nf}\left(\rho V^2\right)_f}{\Delta P_f\left(\rho V^2\right)_{nf}} \tag{32}$$

The total entropy generation (S_T) of the system comprises of the thermal (S_{th}) and viscous (S_{fr}) entropy components. The S_{th} represents the entropy generation supplemented by the heat transfer, whereas S_{fr} denotes the entropy generation due to the viscous dissipation of the system [71].

$$S_T = S_{th} + S_{fr} \tag{33}$$

The values of S_{th} and S_{fr} were determined by the following expressions:

$$S_{th} = \frac{k_{nf}}{T^2}\left[\left(\frac{\partial T}{\partial x}\right)^2 + \left(\frac{\partial T}{\partial y}\right)^2 + \left(\frac{\partial T}{\partial z}\right)^2\right] \tag{34}$$

$$S_{fr} = \frac{\mu_{nf}}{T}\left\{\left[\left(\frac{\partial u_x}{\partial x}\right)^2 + \left(\frac{\partial u_y}{\partial y}\right)^2 + \left(\frac{\partial u_z}{\partial z}\right)^2\right] + \left(\frac{\partial u_x}{\partial y} + \frac{\partial u_y}{\partial x}\right)^2 + \left(\frac{\partial u_x}{\partial z} + \frac{\partial u_z}{\partial x}\right)^2 + \left(\frac{\partial u_z}{\partial y} + \frac{\partial u_y}{\partial z}\right)^2\right\} \tag{35}$$

The global entropy generation rate of the system is calculated by taking the integral of the entropy over the whole computational domain. The global thermal ($S_{th}^{`}$), and viscous ($S_{fr}^{`}$) and total ($S_T^{`}$) entropy generation rate were estimated as follows [71]:

$$S_{th}^{`} = \int \delta_{th} d\mathscr{V} \tag{36}$$

$$S_{fr}^{`} = \int \delta_{fr} d\mathscr{V} \tag{37}$$

$$S_T^{`} = \int \delta_T d\mathscr{V} \tag{38}$$

3. Grid details and model validation

The ICEM CFD package was utilized to discretize the computational domain. The discretization of the solid and fluid computational domains was performed using the structured hexahedral control volumes to accelerate the solution convergence speed. Additionally, a node-to-node association was defined at the interface of the two domains to restrain the interpolation losses. The mesh specifics are detailed in Table 3, and the grid topology of the solid and fluid domains are illustrated in Fig. 1(b).

A grid-independence analysis was performed by considering four grid sizes (N1–N4) of the computational domain to ensure an accurate computational solution insensitive to the mesh resolution. The grid-independence analysis was performed by analyzing the influence of varying grid sizes on the average Nusselt number (Nu_a) and pressure-drop (ΔP) results of the MWCNT/GNPs based (0.25 wt%) hybrid nanofluid for all heat-sink configurations. The Nu_a and ΔP results of the heat-sink configurations demonstrated the least sensitivity towards grid size variation from N3 to N4 with maximum percentage differences of 1.99% and 2.04%, respectively, at Re =

Table 3
Mesh specifics for the grid-independence study on the MWCNT/GNP-H_2O nanofluid-cooled heat sinks with various fin shapes.

Grid Specifics			Hydrofoil		Rhombus		Triangular	
Grid	Δt	N_t	Nu_a	ΔP	Nu_a	ΔP	Nu_a	ΔP
N1	0.232	1,878,38	31.78	1975.02	34.55	1957.0	32.21	2469.7
N2	0.147	3,178,00	25.73	2427.87	28.17	2463.2	27.01	2897.1
N3	0.098	4,237,55	22.01	2619.71	22.74	2795.0	23.50	3124.9
N4	0.056	5,545,67	21.57	2634.19	22.33	2820.5	23.02	3165.4

2000. Therefore, to ensure an accurate solution with the minimum computational cost, all subsequent simulations were performed using grid size N3.

The adopted numerical model was validated through the experimental study of Hussain et al. [23] for the forced convection of the studied nanofluid combinations (MWCNT and MWCNT/GNPs based (0.25 wt%)) in a straight channel subjected to the constant heat-flux boundary condition. Fig. 2 demonstrates that the results for the local heat-transfer coefficient (h_{local}) distribution along the nondimensional stream-wise wall length of the channel for MWCNT-H2O and MWCNT/GNP-H2O nanofluids showed maximum deviations of 6.85% and 5.47%, respectively, with the corresponding experimental results. Meanwhile, for the case of pressure drop, the numerical results showed the maximum percentage errors of 6.95% and 10.14% for the MWCNT-H2O and MWCNT/GNP-H2O

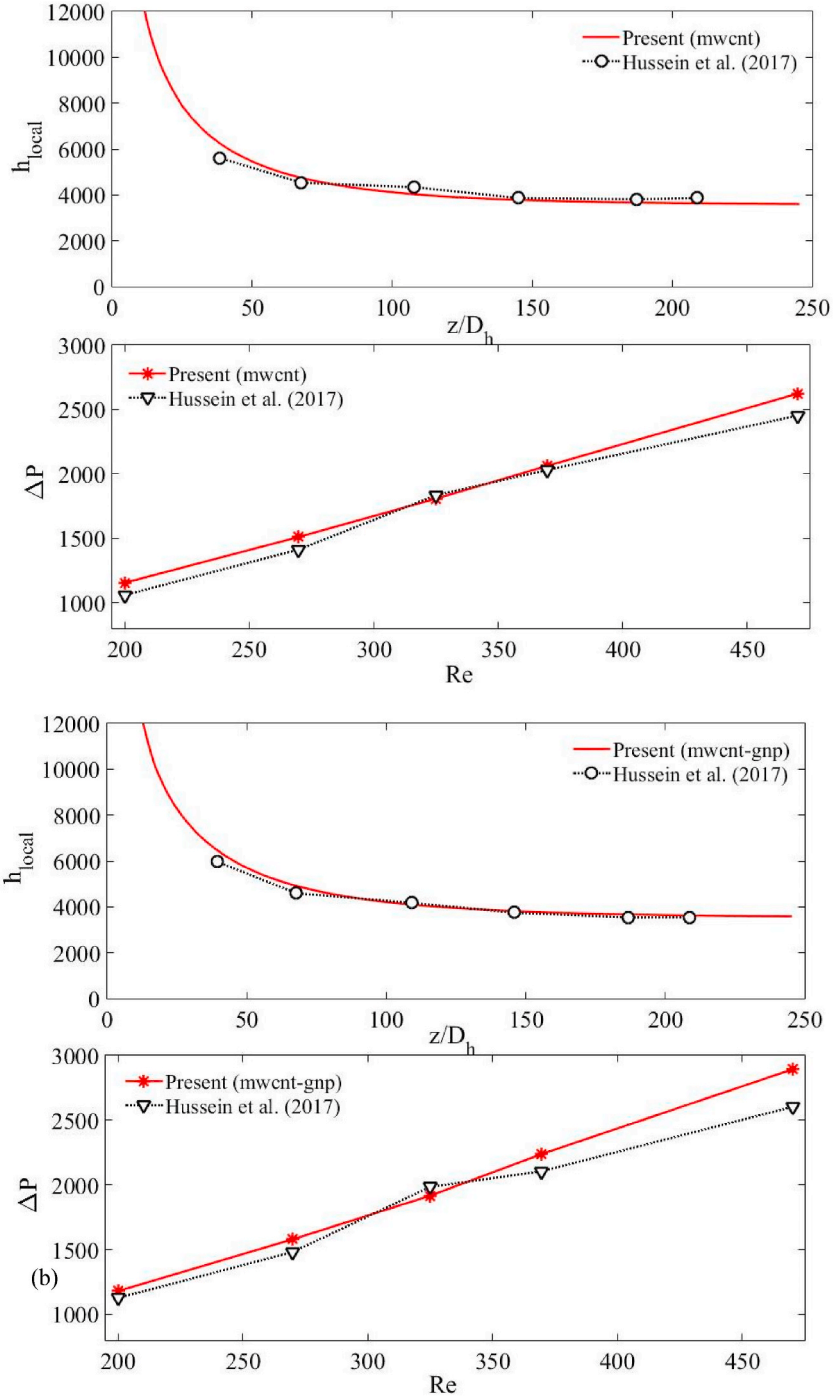


Fig. 2. Validation of the h_{local} distribution and ΔP results of (a) MWCNT-H_2O and (b) MWCNT/GNPs-H_2O nanofluids estimated by the Lagrangian-Eulerian model.

nanofluids, respectively, at Re = 470.

4. Results and discussion

The entropy generation characteristics and hydrothermal performance of MWCNT and MWCNT/GNPs based nanofluids were investigated in a heat sink equipped with the staggered assembly of hydrofoil, rhombus, and triangular pin-fins. The forced convection of the nanofluids with MWCNT and MWCNT/GNP nanoparticles was simulated in the multiphase Lagrangian-Eulerian approximation by incorporating the influence of slip mechanisms and interphase nanolayering. The analysis was performed for the Reynolds number range and nanoparticle weight concentration of $470 \leq Re \leq 2000$ and $\varphi = 0.25\%$, respectively.

4.1. Hydrothermal characteristics

The hydrothermal-performance parameters of the heat sink were estimated in terms of local Nusselt number (Nu_{local}), average Nusselt number (Nu_a), pressure-drop enhancement ratio ($\Delta P_{nf}/\Delta P_f$), and performance-evaluation criteria (PEC).

The variations in average Nusselt number (Nu_a) of water and MWCNT and MWCNT/GNPs based nanofluids depended on Re for the heat sinks of variable pin-fin configurations, as depicted in Fig. 3. The performance of heat sinks was in the descending order of triangular, rhombus, and hydrofoil pin-fin heat sinks irrespective of the coolant type. The addition of nanoparticles to the base fluid resulted in Nu_a improvement at all Re values. However, MWCNT/GNPs based nanofluid considerably outperformed MWCNT nanofluid owing to the additional GNPs concentration in the solution. Irrespective of nanofluid type, the thermal-performance enhancement of the triangular fins was the largest followed by rhombus and hydrofoil fins. The heat transfer improvement of the base fluid with the dispersion of the nanoparticles is generally explained through multiple nanoscale mechanisms in literature such as heat transfer enhancement due to the Brownian motion of the nanoparticles and the corresponding nano-convections of the hosting fluid and thermal-viscous boundary layer disruption, thermophoresis (migration of a nanoparticle in the base fluid due to temperature gradient) and thermal conductivity enhancement of the nanofluids. Most of the studies agree that all these phenomena contribute to heat transfer enhancement of the nanofluids, however, it is not certain which of these phenomena is predominantly involved in heat transfer enhancement of the nanofluids [44,72]. The utilization of MWCNT nanofluid resulted in maximum Nu_a improvements of 13.2%, 18.3%, and 20.7% for hydrofoil, triangular, and rhombus fin heat sinks, respectively. This enhancement further increased to 26.5%, 29.2%, and 30.7%, respectively, by utilizing MWCNT/GNPs based nanofluid.

The local Nusselt number (Nu_{local}) profiles along upstream (xu) and downstream (xd) transverse lines at Re = 2000 is depicted in Fig. 4. xu and xd are plotted at the base of the heat sink in between two upstream (first) and downstream (last) pin-fin rows, respectively. Fig. 4 depict the Nu_{local} profiles of MWCNT and MWCNT/GNPs based nanofluids compared with the water for the heat sinks equipped with hydrofoil, rhombus, and triangular-shaped fins. The Nu_{local} profiles across all heat-sink types showed a roughly sinusoidal profile, with the trough and peaks in the rears and lateral interspacing of the fins, respectively. These peaks and troughs signified the high and low fluid velocity in these regions. The declined coolant velocity in the pin-fin rear indicated flow separation from the fin surface and wake generation behind it. However, the zero values of Nu_{local} profile across hydrofoil and rhombus fins was due to the flow-stagnation points of the succeeding pin-fin rows. Meanwhile, the non-zero Nu_{local} profile values across the triangular fins were due to the relatively larger stream-wise interspacing between the successive pin-fin rows. The amplitude of the Nu_{local} sinusoidal profile gradually increased towards the left of the heat sink probably because of the larger velocity of the incoming fluid. Moreover, for all coolant types, the Nu_{local} at xu was significantly larger than that at xd. This difference signified the reduction in coolant velocity as it flowed across the heat sink owing to the blockage effect prompted by the upstream fins. Considering the effect of

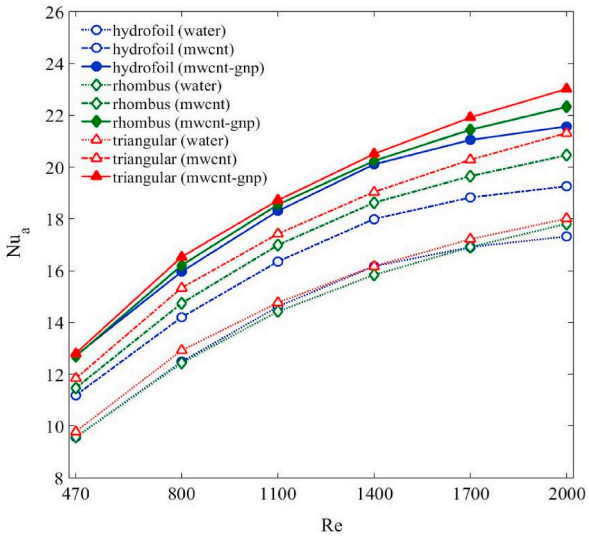


Fig. 3. Average Nusselt number (Nu_a) of nanofluids with Reynolds number (Re).

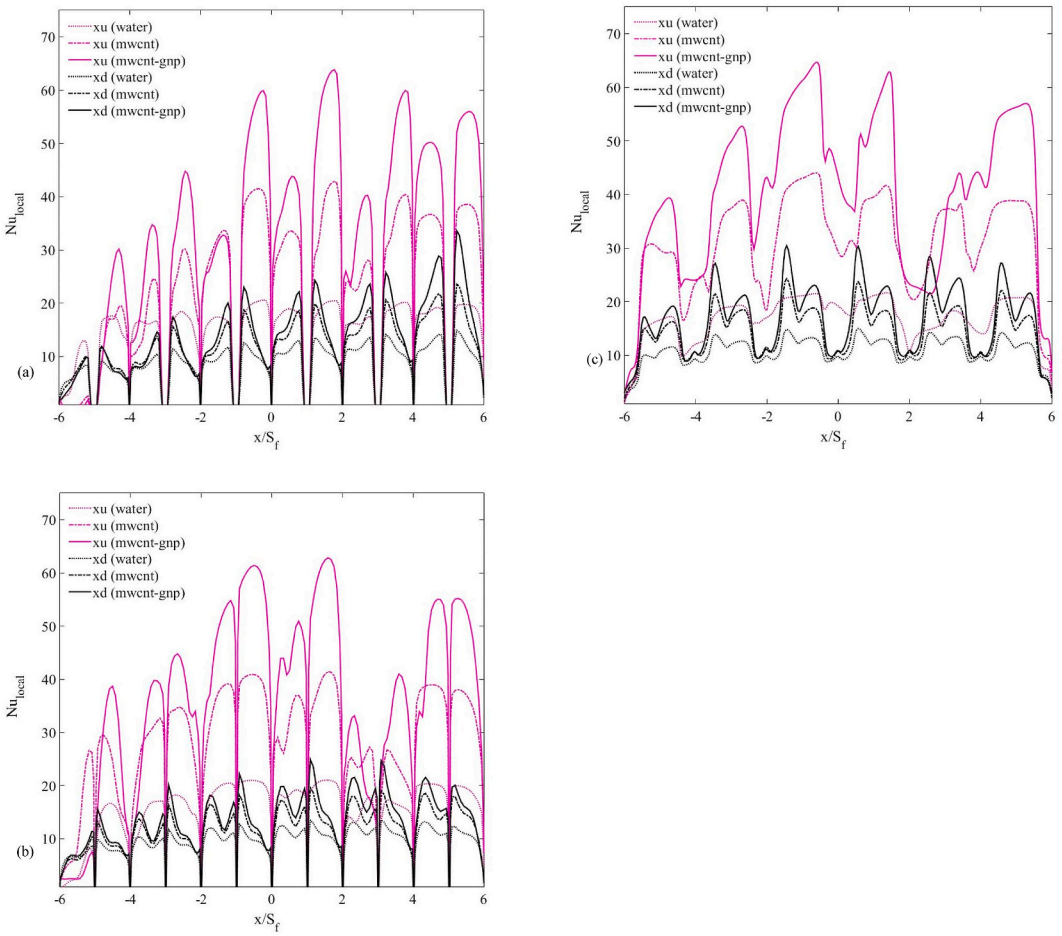


Fig. 4. Local Nusselt number (Nu_{local}) distribution across the xu and xd of the MWCNT/GNPs-H_2O nanofluid-cooled heat sink with (a) hydrofoil, (b) rhombus, and (c) triangular fins at Re = 2000.

different coolant types on Nu_{local}, the utilization of nanofluid resulted in convective heat-transfer enhancement. Moreover, MWCNT/GNPs based outperformed the MWCNT based nanofluid in accordance with the Nu_a trend. Nu_{local} optimization with the nanofluid coolants was most significant in the lateral pin-fin interspacings and across the upstream fins. The Nu_{local} enhancement of the nanofluids gradually decreased as the fluid flowed downstream of the heat sink owing to the decline in interphase interactions at low fluid velocity, i.e., across the xu of the hydrofoil fins, the average Nu_{local} enhancement was noted to be 41% and 85.4% by replacing water with the MWCNT and MWCNT/GNPs based nano-coolants, respectively. However, this Nu_{local} improvement with the utilization of MWCNT and MWCNT/GNPs based nanofluids decreased to 15.2% and 29.7%, respectively, across the downstream fins (xd). Considering different pin-fin types, the Nu_{local} values across the triangular fins were the largest, followed by rhombus and hydrofoil fins. This difference was more significant across the xu of heat sinks. Therefore, the pin-fin efficiencies of the considered pin-fin types were in the descending order of triangular, rhombus, and hydrofoil, respectively, at the largest Re.

The surface streamlines are plotted at an x-y fins bisecting plane of the MWCNT/GNPs based cooled heat sink at Re = 2000 to analyze the coolant distribution in the heat sinks with different pin-fin types. Fig. 5 illustrates that across all heat sinks, the fluid generated a large vortex in the inlet header before its distribution across the pin-fin assembly. Amongst all pin-fin types, this inlet vortex was the strongest for rhombus fins followed by hydrofoil and triangular fins. This inlet vortex explained the non-uniform Nu_{local} profile at xu as it impelled the fluid towards the nearby upstream fins, while the stagnant fluid in the upstream-left recirculation region suppressed the effective heat transfer in this region. The large lateral and small stream-wise interspacing between the hydrofoil and rhombus fins directed the flow towards the diagonal direction across the pin-fin arrangement instead of a straight path due to the streamlined shape of these fins. Such non-uniform flow distribution across these fins and reduced the overall thermal performance of these heat sinks, as previously depicted in Fig. 3. Conversely, the flow was relatively uniformly distributed and stratified across the triangular fins. Moreover, the higher flow velocity across the downstream of the triangular fins explained the largest Nu_{local} values across xd of the respective heat sink (Fig. 4 (c)).

Nanoparticle dispersion improved the heat-dissipation capacity and the pressure drop of the coolant, as depicted in Fig. 6. The $\Delta P_{nf}/\ \Delta P_f$ of MWCNT/GNPs based nanofluid was significantly larger than that of MWCNT based nanofluid obviously because of the

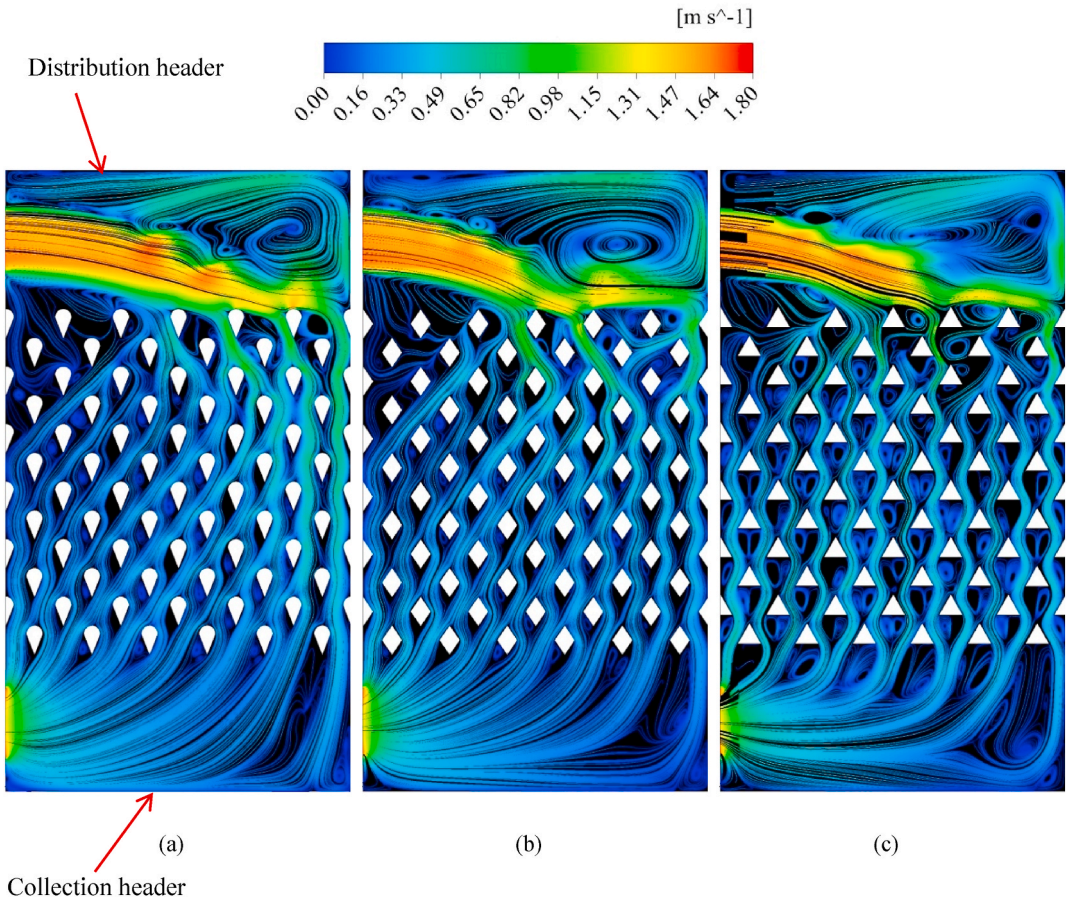


Fig. 5. Flow streamlines across the MWCNT/GNPs-H_2O nanofluid-cooled heat sink with (a) hydrofoil, (b) rhombus, and (c) triangular fins at Re = 2000.

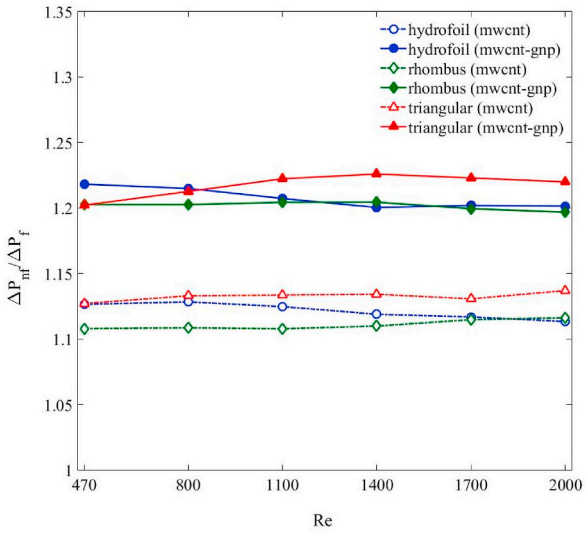


Fig. 6. Pressure drop increment ($\Delta P_{nf}/\Delta P_f$) of nanofluids with Reynolds number (Re).

higher nanoparticle concentration. Irrespective of nanofluid type, the $\Delta P_{nf}/\Delta P_f$ enhancement was the largest for the heat sink with triangular fins, followed by those with hydrofoil and rhombus fins, respectively. At the maximum Re, the utilization of MWCNT/GNPs based nanofluid resulted in pressure-drop augmentations of 13.7%, 11.6%, and 11.3% in the heat sinks with triangular, hydrofoil, and

rhombus fins, respectively. For the case of MWCNT nanofluid, increments of 22%, 20.2%, and 19.7% were noticed for the heat sinks with triangular, hydrofoil, and rhombus fins, respectively, at the maximum Re.

The PEC of the MWCNT and MWCNT/GNPs based nanofluids cooled heat sinks depended on Re, as shown in Fig. 7. Nanoparticle dispersion into the base fluid resulted in PEC improvement of the system at all flow rates. The PEC of different heat sinks was in the descending order of rhombus, triangular, and hydrofoil pin-fin heat sinks irrespective of the coolant type. The PEC of the rhombus pin-fin heat sink outperformed that of the triangular pin-fin heat sink owing to the lower pressure drop. Considering different coolant types, the PEC of MWCNT/GNPs based nanofluids was larger than that of MWCNT nanofluid obviously because of the higher particle concentration. The utilization of MWCNT nanofluid resulted in the maximum PEC improvements of 17.2%, 15.9%, and 9.6% for rhombus, triangular, and hydrofoil fins, respectively. These enhancements increased to 27.1%, 24.5%, and 21%, respectively, upon replacing water coolant with MWCNT/GNPs based nanofluid.

4.2. Entropy generation characteristics

The hydrothermal performance of a system can also be evaluated by using entropy generation as a measure of the relative merit of heat transfer augmentation techniques. The favorable operational and design conditions are those which are capable of reducing the entropy (irreversibility) of the useful energy (exergy) [73]. The second law of thermodynamics signifies the extent of the irreversibility of a process. The entropy generation aspects of MWCNT and MWCNT/GNPs based nanofluids in heat sinks with various pin-fin configurations were evaluated in terms of the global total ($S_T^{'}$), global thermal ($S_{th}^{'}$), and global viscous ($S_{fr}^{'}$) entropy generation rate of the system.

Fig. 8 (a, b) depicts the variation of global thermal ($S_{th}^{'}$) and viscous ($S_{fr}^{'}$) entropy generation rate with the Re for water, MWCNT, and MWCNT/GNPs based nanofluids in heat sinks with hydrofoil, rhombus, and triangular fins. The $S_{th}^{'}$ of all the coolant types illustrated an inverse association with Re, in agreement to general trends. The addition of nanoparticles into host fluid led to an adverse effect on $S_{th}^{'}$ for all heat-sink types. Therefore, the $S_{th}^{'}$ of different thermal fluids was in the descending order of water, MWCNT and MWCNT/GNPs based nanofluids. The reduction in $S_{th}^{'}$ with the addition of nanoparticles was more significant at smaller Re and larger φ. It is because the larger φ and Re intensify the fluid mixing and interphase interactions, thereby resulting in a temperature-gradient reduction inside the heat sink. Consequently, $S_{th}^{'}$ decreased because it depended on the average temperature and spatial gradient of temperature (Eq. (34)). Irrespective of thermal fluid type, the $S_{th}^{'}$ of different heat sinks were in the descending order of hydrofoil, rhombus, and triangular pin-fin heat sinks at the maximum Re, in agreement to previously stated results. At the largest Re, the utilization of MWCNT nanofluids resulted in 53.17%, 60.40%, and 40.26% $S_{th}^{'}$ reduction for the triangular, rhombus, and hydrofoil pin-fin heat sinks, respectively. Meanwhile, the percentage difference between the $S_{th}^{'}$ values of water and MWCNT/GNPs based nanofluid-cooled heat sinks were noted to be 76.86%, 78.75%, and 73.61%, respectively, at the largest Re. For the case of δP_{fr}, as the addition of the nanoparticles into the base fluid results in viscosity augmentation, the $S_{fr}^{'}$ of MWCNT and MWCNT/GNPs based nanofluids were larger than that of water for all heat-sink types and Re. In accordance with the pressure-drop results, $S_{fr}^{'}$ of different heat sinks were in the descending order of triangular, hydrofoil, and rhombus pin-fin heat sinks. The utilization of MWCNT nanofluid resulted in the maximum $S_{fr}^{'}$ increments of 19.68%, 15.91%, and 14.32% for triangular, hydrofoil, and rhombus fins, respectively. These augmentations in $S_{fr}^{'}$ were noted to be 33.67%, 29.29%, and 27.62%, respectively, upon replacing water coolant with MWCNT/GNPs based nanofluid.

Fig. 9 shows the total entropy generation rate ($S_T^{'}$) of different heat sinks cooled by water and MWCNT and MWCNT/GNPs based

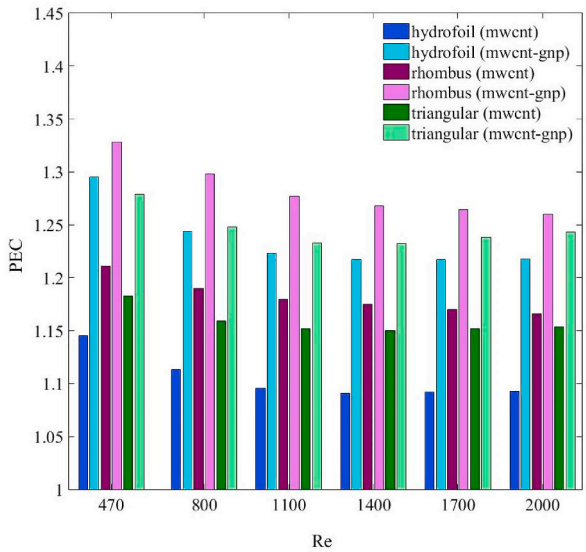


Fig. 7. Performance evaluation criteria (PEC) of nanofluids with Reynolds number (Re).

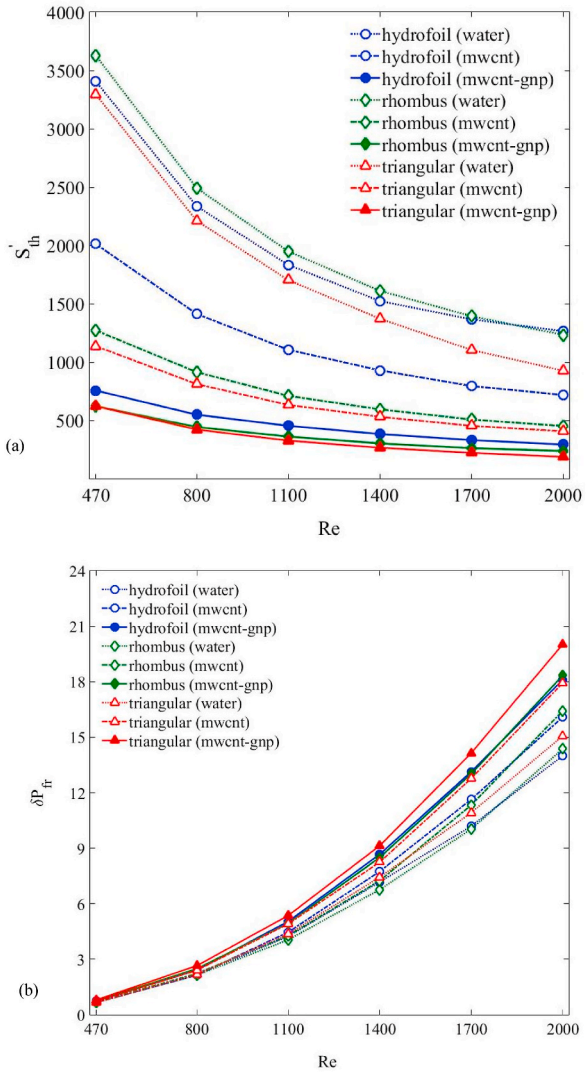


Fig. 8. Variation of (a) global thermal ($S_{th}^{'}$) and (b) viscous ($S_{fr}^{'}$) entropy generation rate of water and nanofluids with Reynolds number ().

nanofluids as a function of Re. Results revealed that the trend of $S_T^{'}$ and its dependence on Re and φ were similar to those of $S_{th}^{'}$ because the total entropy generation rate of the system was an algebraic sum of $S_{th}^{'}$ and $S_{fr}^{'}$ (Eq. (38)). In the present cases, the contribution of $S_{th}^{'}$ was considerably higher than that of $S_{fr}^{'}$ in determining $S_T^{'}$ (Fig. 8).

The contours of thermal entropy generation rate ($S_{th}^{'}$) across the MWCNT/GNPs-H_2O nanofluid-cooled heat sink (x-y fins bisecting plane) with variable pin-fin heat sink types are shown in Fig. 10. In agreement with Fig. 8, $S_{th}^{'}$ of the hydrofoil pin-fin heat sink was largest followed by rhombus and triangular pin-fin heat sinks.

5. Conclusion

The entropy generation characteristics and hydrothermal performance of MWCNT and MWCNT/GNPs-H_2O nanofluid are investigated in a heat sink equipped with the staggered assembly of hydrofoil, rhombus, and triangular fins. The forced convection of the nanofluid was simulated through the multiphase Eulerian-Lagrangian model (discrete-phase model) by including the influence of slip mechanisms (Brownian motion, thermophoresis, Saffman lift and drag forces, pressure forces, virtual mass, and gravity) and the interphase nanolayering effect. The analysis was performed for the Reynolds number range and nanoparticle weight concentration of $470 \leq Re \leq 2000$ and $\varphi = 0.25\%$, respectively. The conclusions drawn were as follows.

- The thermal-performance enhancement of the nanofluids in the heat sinks depended on the pin-fin shape. The performance of the MWCNT-H_2O and MWCNT/GNPs-H_2O nanofluids in different heat sinks were in the descending order of triangular, rhombus, and hydrofoil pin-fin heat sink.

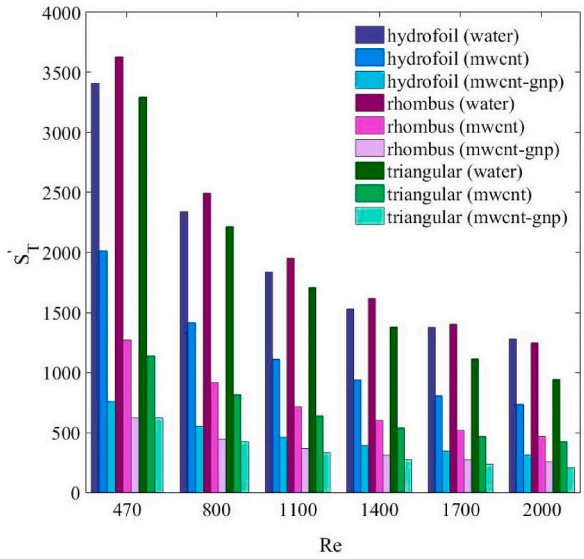


Fig. 9. Variation in global total entropy generation rate (S_{th}) of water and nanofluids with Reynolds number (Re).

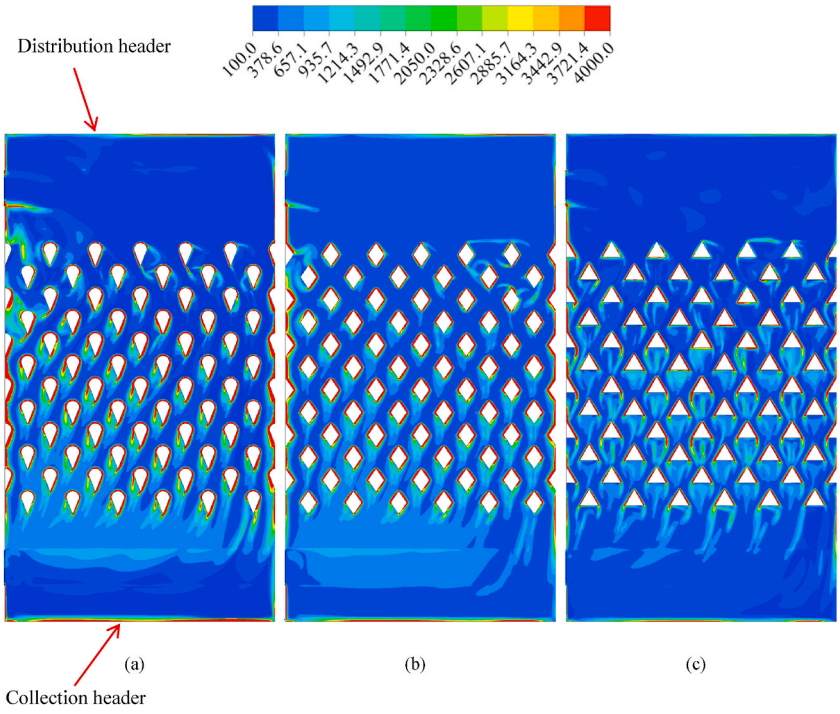


Fig. 10. Thermal entropy generation rate contours across the MWCNT/GNPs-H_2O nanofluid-cooled heat sink with (a) hydrofoil, (b) rhombus, and (c) triangular fins at Re = 2000.

- The streamlined shape along with the large lateral and small stream-wise interspacing between the hydrofoil and rhombus fins resulted in poor coolant distribution across the heat sinks, thereby deteriorating their heat-dissipation capacity.
- The pressure-drop augmentation of both nanofluids was the largest for the heat sink with triangular fins, followed by those with hydrofoil and rhombus fins, respectively.
- The PEC of different heat sinks was in the descending order of rhombus, triangular, and hydrofoil pin-fin heat sinks irrespective of coolant type. The PEC of the rhombus pin-fin heat sink outperformed that of the triangular pin-fin heat sink owing to the lower pressure drop in the former.

- Across all heat-sink types, MWCNT/GNPs-H_2O nanofluids significantly outperformed MWCNT nanofluids owing to the additional concentration of highly thermal conductive GNPs.

Author declaration

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

Authorship contribution

Tehmina Ambreen: Conceptualization, Methodology, Analysis, Data Interpretation and Writing Original Draft.
Arslan Saleem: Meshing and Modelling.
Muhammad Tanveer: Writing, Reviewing and Editing.
Anirudh Kulkarni: Reviewing and Editing.
Cheol Woo Park: Reviewing and Editing, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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